VoteTrace360

Frontend Developer API Reference

Base URL	http://<baseURL> (https once we set up ssl)
Version	1.0 — March 2026
Auth	None — all endpoints are open, no authentication required
Gateway	All traffic goes through NGINX on port 80 — never call microservice ports directly, or 443 with ssl
Data Format	JSON for all requests and responses. Amounts are in Kenyan Shillings (KES).

This document contains everything you need to build the frontend. Each endpoint includes the full URL to call, what to send (if anything), and a sample response with field-by-field explanations. The physical assets endpoint is pending and will be added when ready.

How to Call an Endpoint

Every request goes to the base URL followed by the endpoint path. No ports, no service names — just localhost through NGINX.

URL Format

```
Base URL:  http://<baseURL>

Pattern:   http://<baseURL>{endpoint}

Examples:
  http://<baseURL>/api/v1/candidates/
  http://<baseURL>/api/v1/candidates/5/summary/
  http://<baseURL>/api/v1/digital/3
  http://<baseURL>/api/v1/donors/7
```

Making a GET request (JavaScript fetch)

```
// List all candidates
const res = await fetch('http://<baseURL>/api/v1/candidates/');
const candidates = await res.json();
```

```
// Get one candidate summary
const res = await fetch('http://<baseURL>/api/v1/candidates/5/summary/');
const summary = await res.json();

// Get digital spend for candidate 3
const res = await fetch('http://<baseURL>/api/v1/digital/3');
const digital = await res.json();
```

Making a POST request (JavaScript fetch)

```
// Add a single donor record
const res = await fetch('http://<baseURL>/api/v1/donors/record', {
  method: 'POST',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify({
    candidate_id: 1,
    donor_name: 'Nairobi Holdings Ltd',
    donation_amount: 500000,
    registration_date: '2024-12-01',
    donor_type: 'company',
    donation_date: '2025-03-01'
  })
});
const result = await res.json();
```

1. All Endpoints at a Glance

Use this as your quick reference while building components.

Method	Full URL	Use For
GET	`http://<baseURL>/health/reconciliation/`	Confirm reconciliation service is up
GET	`http://<baseURL>/health/digital/`	Confirm digital spend service is up
GET	`http://<baseURL>/health/donors/`	Confirm donor service is up
GET	`http://<baseURL>/api/v1/candidates/`	Candidate listing page — all 10 candidates with scores
GET	`http://<baseURL>/api/v1/candidates/:id/summary/`	Candidate detail page — full breakdown for one candidate
GET	`http://<baseURL>/api/v1/digital/:candidateId`	Digital spend charts — platform breakdown
POST	`http://<baseURL>/api/v1/digital/import`	Bulk import digital spend from CSV
POST	`http://<baseURL>/api/v1/digital/record`	Add one digital spend record live

GET	http://<baseURL>/api/v1/donors/:candidateId	Donor risk table — all donors with risk scores
POST	http://<baseURL>/api/v1/donors/upload	Bulk import donors from CSV
POST	http://<baseURL>/api/v1/donors/record	Add one donor record live

2. Health Checks

Run these in the browser before starting development. If any return an error, the corresponding service is down.

GET `http://<baseURL>/health/reconciliation/`

Reconciliation Engine — candidate scores

Response

```
{
  "status": "ok",
  "service": "reconciliation-engine",
  "timestamp": "2025-03-04T14:30:00.000Z"
}
```

GET `http://<baseURL>/health/digital/`

Digital Spend Service

Response

```
{
  "status": "ok",
  "service": "reconciliation-engine",
  "timestamp": "2025-03-04T14:30:00.000Z"
}
```

GET `http://<baseURL>/health/donors/`

Donor & Risk Service

Response

```
{
  "status": "ok",
  "service": "reconciliation-engine",
  "timestamp": "2025-03-04T14:30:00.000Z"
}
```

3. Candidate Endpoints

Served by the Reconciliation Engine. These are the two most important endpoints — almost every page in the app will call one of these.

GET `http://<baseURL>/api/v1/candidates/`

Returns all 10 candidates sorted by integrity score ascending (most suspicious first). Use this to build the candidate listing/cards page.

Response — Array of candidate objects

```
[
  {
    "id": 5,
    "name": "Mr. Kipchoge Ruto",
    "party": "Independent",
    "position": "Member of Parliament",
    "constituency": "Uasin Gishu",
    "integrity": {
      "score": 0.21,
      "risk_level": "RED",
      "risk_label": "High Risk / Dark Money Signal"
    },
    "financial_summary": {
      "total_physical_spend": 4680000,
      "total_digital_spend": 14500000,
      "total_estimated_spend": 19180000,
      "total_reported_income": 23500000,
      "spending_gap": 0,
      "spending_gap_ratio": 1.0
    },
    "updated_at": "2025-03-04T14:30:00Z"
  },
  { ...9 more candidates }
]
```

Field Reference

Field	Value / Description
`id`	Numeric candidate ID (1-10). Use this in all other endpoint URLs.
`name`	Full name of the candidate.
`party`	Political party affiliation.
`position`	Elected position being contested e.g. Governor, Senator, MP.

`constituency`	County or constituency name.
`integrity.score`	Number between 0.00 and 1.00. Higher = more transparent.
`integrity.risk_level`	GREEN, AMBER, or RED. Use this for badge colour.
`integrity.risk_label`	Human-readable label. Use as tooltip or sub-label on badge.
`financial_summary.total_physical_spend`	KES — sum of all physical asset costs (billboards, rallies, etc).
`financial_summary.total_digital_spend`	KES — sum of all digital ad spend (Meta, Google, X).
`financial_summary.total_estimated_spend`	KES — physical + digital combined.
`financial_summary.total_reported_income`	KES — sum of all donor declarations filed.
`financial_summary.spending_gap`	KES — estimated spend minus reported income. 0 means clean or data pending.
`financial_summary.spending_gap_ratio`	estimated_spend / reported_income. Values > 1 mean spending exceeds declared income.
`updated_at`	ISO timestamp of last reconciliation run for this candidate.

GET `http://<baseURL>/api/v1/candidates/:id/summary/`

Full integrity breakdown for one candidate. Use this to build the individual candidate detail/dashboard page. Replace :id with the candidate's numeric ID.

Example URLs

```
http://<baseURL>/api/v1/candidates/1/summary/    →  Hon. Wanjiku Kamau   (GREEN)
http://<baseURL>/api/v1/candidates/5/summary/    →  Mr. Kipchoge Ruto    (RED)
http://<baseURL>/api/v1/candidates/3/summary/    →  Eng. Hassan Abdi     (AMBER)
```

Response


```
{
  "candidate": {
    "id": 5,
    "name": "Mr. Kipchoge Ruto",
    "party": "Independent",
    "position": "Member of Parliament",
    "constituency": "Uasin Gishu"
  },
  "integrity": {
    "score": 0.21,
    "risk_level": "RED",
```

```
    "classification": "High Risk / Dark Money Signal"
  },
  "financial_summary": {
    "total_physical_spend": 4680000,
    "total_digital_spend": 14500000,
    "total_estimated_spend": 19180000,
    "total_reported_income": 23500000,
    "spending_gap": 0,
    "spending_gap_ratio": 1.0,
    "unreported_percentage": 0
  },
  "donor_risk": {
    "avg_donor_risk": 0.9
  },
  "updated_at": "2025-03-04T14:30:00Z"
}
```

Additional Field Reference

Field	Value / Description
`integrity.classification`	Same as risk_label in the list endpoint. Show on the detail page banner.
`financial_summary.unreported_percentage`	Percentage of estimated spend that has no corresponding donor declaration. Use for a "dark money %" stat.
`donor_risk.avg_donor_risk`	Average risk score across all donors (0.0 = all clean, 1.0 = all HIGH risk). Use for a donor trust gauge/meter.

4. Digital Spend Endpoints

Served by the Digital Spend Service. Use these for ad spend charts and platform breakdowns.

GET `http://<baseURL>/api/v1/digital/:candidateId`

Digital advertising spend for a candidate, aggregated by platform. Use for bar charts, pie charts, and spend comparison UI.

Example URLs

```
http://<baseURL>/api/v1/digital/1    →  Hon. Wanjiku Kamau
http://<baseURL>/api/v1/digital/5    →  Mr. Kipchoge Ruto (high spend)
```

Response

```
{
  "candidate_id": 1,
  "total_digital_spend": 2500000,
```

```
  "platform_breakdown": [
    {
      "platform": "meta",
      "spend_amount": 1250000,
      "percentage": 50.0,
      "campaign_count": 3,
      "period_start": "2025-01-01",
      "period_end": "2025-02-28"
    },
    { "platform": "google", "spend_amount": 875000,  "percentage": 35.0,
"campaign_count": 3 },
    { "platform": "x",      "spend_amount": 375000,  "percentage": 15.0,
"campaign_count": 3 }
  ],
  "records": [ ...raw individual records array... ]
}
```

Field Reference

Field	Value / Description
`total_digital_spend`	Total KES spent on digital ads across all platforms.
`platform_breakdown`	Array — one object per platform (meta, google, x).
`platform`	Platform name: "meta", "google", or "x".
`spend_amount`	KES spent on this platform.
`percentage`	Share of total digital spend for this platform. Use for pie chart segments.
`campaign_count`	Number of separate ad campaigns recorded for this platform.
`records`	Raw individual records. Use if you need granular timeline data.

POST `http://<baseURL>/api/v1/digital/import`

Bulk import digital spend records from a CSV file. Triggers automatic score recalculation for all affected candidates.

Request — Postman Setup

Field	Value / Description
`Method`	POST
`URL`	http://<baseURL>/api/v1/digital/import
`Body type`	form-data
`Field name`	file (must be exactly this name)

`Field value`	Select the CSV file from your computer

CSV Required Columns

Field	Value / Description
`candidate_id`	Numeric candidate ID (1-10)
`platform`	meta | google | x
`spend_amount`	Amount in KES as a plain number e.g. 150000
`period_start`	Start date of the ad campaign: YYYY-MM-DD
`period_end`	End date of the ad campaign: YYYY-MM-DD

Response

```
{
  "success": true,
  "message": "Successfully imported 30 digital spend records",
  "rows_imported": 30,
  "candidates_affected": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
}
```

POST `http://<baseURL>/api/v1/digital/record`

Add a single digital spend record. Use this for live updates during the demo — when a new ad is detected, call this and the candidate score updates immediately.

Request Body (JSON)

```
{
  "candidate_id": 1,
  "platform": "meta",
  "spend_amount": 150000,
  "period_start": "2025-03-01",
  "period_end": "2025-03-07"
}
```

Response

```
{
  "success": true,
  "message": "Digital spend record added and reconciliation triggered",
  "record": {
    "id": 31,
    "candidate_id": 1,
    "platform": "meta",
    "spend_amount": 150000,
```

```
    "period_start": "2025-03-01",
    "period_end": "2025-03-07",
    "created_at": "2025-03-04T14:30:00Z"
  }
}
```

5. Donor & Risk Endpoints

Served by the Donor & Risk Service. Risk is scored automatically — you just display what comes back.

GET `http://<baseURL>/api/v1/donors/:candidateId`

All donors for a candidate with risk scores, red flags, and concentration analysis. Use for the donor risk table and red flag summary cards.

Example URLs

```
http://<baseURL>/api/v1/donors/1    →  Hon. Wanjiku Kamau (all LOW risk donors)
http://<baseURL>/api/v1/donors/5    →  Mr. Kipchoge Ruto  (4 HIGH risk donors)
```

Response

```
{
  "candidate_id": 5,
  "total_reported_income": 23500000,
  "donor_count": 5,
  "risk_summary": {
    "high": 4,
    "medium": 0,
    "low": 1,
    "high_risk_percentage": 80.0
  },
  "red_flags": {
    "briefcase_companies": 3,
    "unverifiable_entities": 1,
    "concentration_risk": {
      "top_donor_percentage": 34.0,
      "top_3_donors_percentage": 78.7
    }
  },
  "donors": [
    {
      "id": 21,
      "donor_name": "Skyline Enterprises Ltd",
      "donation_amount": 8000000,
      "registration_date": "2024-11-20",
      "company_age_days": 56,
      "risk_score": "HIGH",
      "risk_factors": ["Company registered < 90 days before donation"],
      "donation_percentage": 34.0
    },
    {
      "id": 24,
      "donor_name": "Mombasa Traders Co",
```

```
      "donation_amount": 4500000,
      "registration_date": null,
      "company_age_days": null,
      "risk_score": "HIGH",
      "risk_factors": ["No company record found in Registrar of Companies"],
      "donation_percentage": 19.1
    }
  ]
}
```


Field Reference

Field	Value / Description
`total_reported_income`	Total KES across all donor declarations for this candidate.
`risk_summary.high`	Count of donors flagged as HIGH risk.
`risk_summary.high_risk_percentage`	Percentage of donors that are HIGH risk. Use for a risk gauge.
`red_flags.briefcase_companies`	Count of companies registered < 90 days before donating.
`red_flags.unverifiable_entities`	Count of companies with no registration record found.
`concentration_risk.top_donor_percentage`	% of total income from the single largest donor.
`concentration_risk.top_3_donors_percentage`	% of total income from the top 3 donors combined.
`donors[].risk_score`	HIGH, MEDIUM, or LOW. Use for row colour in the donor table.
`donors[].risk_factors`	Array of strings explaining why the donor was flagged. Show as tooltips or expand rows.
`donors[].company_age_days`	How old the company was at time of donation. null means individual or unverifiable.
`donors[].donation_percentage`	% share of total income this donor represents.

POST `http://<baseURL>/api/v1/donors/upload`

Bulk import donor records from a CSV file. Risk scoring is automatic. Triggers score recalculation for all affected candidates.

Request — Postman Setup

Field	Value / Description

Method	POST
URL	http://<baseURL>/api/v1/donors/upload
Body type	form-data
Field name	file (must be exactly this name)
Field value	Select the CSV file from your computer

CSV Required Columns

Field	Value / Description
candidate_id	Numeric candidate ID (1-10)
donor_name	Full name of the donor company or individual
donation_amount	Amount in KES as a plain number e.g. 3000000
registration_date	Company registration date: YYYY-MM-DD. Leave blank for individuals.
donor_type	company | individual
donation_date	Date the donation was made: YYYY-MM-DD. Used to calculate company age.

Response

```
{
  "success": true,
  "message": "Successfully imported 57 donor records",
  "rows_imported": 57,
  "candidates_affected": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
  "risk_summary": { "total": 57, "high": 15, "medium": 9, "low": 33 }
}
```

POST **http://<baseURL>/api/v1/donors/record**

Add a single donor record. Risk is scored immediately against existing donors for that candidate. Score recalculates instantly.

Request Body (JSON)

```
{
  "candidate_id": 1,
  "donor_name": "Nairobi Holdings Ltd",
  "donation_amount": 500000,
  "registration_date": "2024-12-01",
  "donor_type": "company",
  "donation_date": "2025-03-01"
```

```
}
```

ℹ *registration_date and donation_date are optional but strongly recommended — without them, risk scoring is less accurate.*

Response

```
{
  "success": true,
  "message": "Donor record added and reconciliation triggered",
  "record": {
    "id": 58,
    "candidate_id": 1,
    "donor_name": "Nairobi Holdings Ltd",
    "donation_amount": 500000,
    "company_age_days": 91,
    "risk_score": "LOW",
    "risk_factors": [],
    "donation_percentage": 5.8
  }
}
```

6. Error Responses

All services return consistent error shapes. Handle these in your frontend to show appropriate messages to users.

Status	When It Happens	Response Body
400	Non-numeric ID in URL	`{"error": "candidateId must be a number"}`
404	Candidate not found or scores not calculated yet	`{"error": "No summary found for candidate 99"}`
422	CSV row failed validation on import	`{"success": false, "errors": [{"row": 3, "error": "platform must be one of: meta, google, x"}]}`
500	Supabase or internal server error	`{"error": "Internal server error", "detail": "..."}`

7. Risk Level UI Reference

Use these values from the API to drive badge colours, row highlights, and status indicators across the UI.

risk_level	Score Range	risk_label	Suggested Colour	What It Means
GREEN	0.80 – 1.00	Transparent	`#10B981 / green-500`	Declared income covers estimated spend. Donors are credible.
AMBER	0.50 – 0.79	Requires Review	`#F59E0B / amber-500`	Moderate gap or some suspicious donors. Warrants scrutiny.

RED	0.00 – 0.49	High Risk / Dark Money	`#EF4444 / red-500`	Large unexplained spend or majority high-risk donors.

Donor risk_score colours

risk_score	Suggested Row/Badge Colour	Trigger Condition
HIGH	`#FEE2E2 row / #EF4444 badge`	Company < 90 days old, or no registration record found
MEDIUM	`#FEF3C7 row / #F59E0B badge`	Single donor contributes > 40% of total reported income
LOW	`#D1FAE5 row / #10B981 badge`	No flags triggered — legitimate company or individual donor

8. Demo Candidates Quick Reference

These are the 10 seeded candidates in the database. Use the IDs in your API calls during development.

ID	Name	Party	Position	County	Profile
1	Hon. Wanjiku Kamau	UDA	Governor	Nairobi	**GREEN**
2	Dr. Otieno Ochieng	ODM	Governor	Kisumu	**GREEN**
3	Eng. Hassan Abdi	Jubilee	Senator	Mombasa	**AMBER**
4	Ms. Njeri Mwangi	Wiper	Woman Rep	Kiambu	**GREEN**
5	Mr. Kipchoge Ruto	Independent	MP	Uasin Gishu	**RED**
6	Prof. Amina Shariff	ANC	Governor	Garissa	**AMBER**
7	Hon. Mwenda Kariuki	UDA	Senator	Kirinyaga	**RED**
8	Mr. Omondi Adhiambo	ODM	MP	Homa Bay	**AMBER**
9	Dr. Fatuma Noor	Jubilee	Woman Rep	Isiolo	**GREEN**
10	Mr. Baraka Mwamba	Independent	Governor	Kwale	**RED**